founding “an anti-Soviet group,” was imprisoned for six years, and then after seven years of freedom was rearrested and sent into five years’ exile in the Komi Republic, 1200 kilometers east of Leningrad.

I have never met Aleksandrov or Pimenov, but tales of Pimenov were still rippling through Leningrad’s community of scientists when I visited there in 1971, a year after Pimenov’s second arrest. Rumor had it that Pimenov viewed the Soviet government as morally corrupt, and, like many young people in America during the Vietnam War, he felt that, if he cooperated with the government, the government’s corruption would rub off on him. The only way to feel morally clean was through civil disobedience. In America, civil disobedience meant refusing to register for the draft. For Pimenov, civil disobedience meant *samizdat*. Samizdat was the “self-publication” of forbidden manuscripts. Pimenov, it was rumored, would receive from friends a manuscript which had been forbidden for publication in the Soviet Union, he would type out a half-dozen copies using carbon paper, and he would then pass those copies on to other friends, who would repeat the process. Pimenov got caught, was convicted, and was sentenced to five years’ exile in the Komi Republic, where he worked as a tree-feller and an electrician in a sawmill until the Komi Academy of Sciences took advantage of his exile and made him the chair of their mathematics department.

Finally able to do mathematics again, Pimenov continued his topological studies of spacetime. By then topology had taken firm root as a key tool for physicists’ gravitation research, but Pimenov remained isolated from the leading physicists of his country. He never had the impact that, under other circumstances, he might have.

Roger Penrose, by contrast with Aleksandrov and Pimenov, lives with one foot firmly planted in the mathematics community and the other firmly planted in physics, and this has been a major source of his success.

Best Guesses

One might have thought that Penrose’s singularity theorem would settle once and for all the question of what is inside a black hole. Not so. Instead it opened up a new set of questions—questions with which physicists have